



E.5 Fusion and stars

Nuclear and quantum physics · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Nuclear fusion is the combination of two light nuclei (such as isotopes of hydrogen) into a heavier nucleus, releasing energy because the product nucleus has greater binding energy per nucleon than the reactants, with the mass defect converted to energy via $E = \Delta mc^2$; fusion requires extremely high temperatures and densities to overcome electrostatic (Coulomb) repulsion between positively charged nuclei so they can approach close enough for the strong nuclear force to bind them. Stars are powered by sustained nuclear fusion (mainly hydrogen fusing into helium) in their cores, which provides an outward radiation/thermal pressure that balances the inward pull of gravity (hydrostatic equilibrium); a star's luminosity L , surface temperature T and radius r are related by $L = 4\pi r^2 \sigma T^4$ (Stefan-Boltzmann law), and stars occupy characteristic regions on the Hertzsprung-Russell (HR) diagram (luminosity vs. surface temperature), from which their evolutionary stage and eventual fate (white dwarf, neutron star, or black hole) can be inferred based on mass.

Worked example: follow the reasoning

A fusion reaction converts a mass defect of 6.0×10^{-29} kg into energy. Find the energy released. ($c = 3.00 \times 10^8$ m·s⁻¹)
 $E = \Delta mc^2 = (6.0 \times 10^{-29})(3.00 \times 10^8)^2 = (6.0 \times 10^{-29})(9.00 \times 10^{16}) = 5.4 \times 10^{-12}$ J.
 Answer: $E = 5.4 \times 10^{-12}$ J.

Quick check · multiple choice

Why does nuclear fusion require extremely high temperatures to occur?
 A To provide enough thermal energy for gamma rays to be released
 B To give nuclei enough kinetic energy to overcome their mutual electrostatic (Coulomb) repulsion
 C To increase the number of neutrons available
 D To slow the nuclei down so the strong force has time to act

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A main-sequence star has luminosity 3.0×10^{26} W and surface temperature 5500 K. ($\sigma = 5.67 \times 10^{-8}$ W·m⁻²·K⁻⁴) [7]
 - (a) Using $L = 4\pi r^2 \sigma T^4$, calculate the radius of the star.
 - (b) State what physical balance (hydrostatic equilibrium) keeps a main-sequence star stable in size, and identify the two opposing effects involved.
 - (c) Explain what will eventually happen to the balance in part (b) once the star exhausts the hydrogen fuel in its core, in general terms.

Final answer: _____



Answers and explanation

Question 1 [7]

(a) $r^2 = L/(4\pi\sigma T^4) = 3.0 \times 10^{26}/[4\pi(5.67 \times 10^{-8})(5500^4)]$.

$5500^4 = 9.15 \times 10^{14}$ (approx). Denominator = $4\pi(5.67 \times 10^{-8})(9.15 \times 10^{14}) = 4\pi(5.19 \times 10^7) = 6.52 \times 10^8$.

$r^2 = 3.0 \times 10^{26}/6.52 \times 10^8 = 4.60 \times 10^{17} \rightarrow r = 6.79 \times 10^8 \text{ m}$.

(b) Hydrostatic equilibrium is the balance between the inward gravitational force (tending to collapse the star) and the outward thermal/radiation pressure generated by fusion reactions in the core (tending to expand the star); these two effects balance to keep the star's radius stable.

(c) Once core hydrogen is exhausted, fusion energy generation in the core decreases, so outward pressure can no longer balance gravity; the core contracts and heats up, which can ignite further fusion stages (e.g. helium fusion) and cause the outer layers to expand, moving the star off the main sequence (e.g. becoming a red giant), with its ultimate fate then depending on its mass.

Quick check answer

B. Nuclei are positively charged and repel each other electrostatically; only at very high temperatures do nuclei have enough kinetic energy to get close enough for the short-range strong nuclear force to overcome this repulsion and bind them together.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Physics HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.